ISTE240-Group9 Project Documentation

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Semester: Spring

Year: 2025

Course: ISTE-240.600

April 26, 2026

Rubric

FRONTEND (20 pts) – CLOS 2, 3 and 5

18-20: Professional + Bootstrap + responsive + toasts + loading states

14-17: Clean + responsive + basic response messages

10-13: Basic CSS + not responsive

0-9: No CSS or broken

REST API (25 pts) – CLOS 4 and 5

22-25: Each student has own controller + All 6 methods

17-21: Each has controller + 4-5 methods

12-16: Only 1-2 controller + 2-3 methods

0-11: Only one controller + Less than 2 methods

JPA (20 pts) – CLO 4

17-20: Each student has own repository + 5+ methods + entity classes

13-16: Each has repository + 3-4 methods

9-12: Only 1-2 repository + 1-2 methods

0-8: One repository for all → ZERO for section

GIT (10 pts) – CLO 1

9-10: Branches + 5+ commits + good messages + PRs

7-8: Branches + 3-4 commits

5-6: One branch + 1-3 commits

0-4: Single commit

DEMO (15 pts) – CLOS 2, 3 and 5

13-15: 5+ pages + all CRUD + no errors

10-12: 4-5 pages + most CRUD

7-9: 3 pages + basic CRUD

0-6: Less than 3 pages

FINAL REPORT (10 pts) – CLO 1

9-10: Complete documentation + API docs + team contributions + screenshots

7-8: Most sections present + basic API docs

5-6: Basic report + missing key sections

0-4: No report or very incomplete

FINAL PRESENTATION (10 pts) – CLO 1

9-10: Clear explanation + live demo works + answers questions confidently

7-8: Good presentation + minor demo issues

5-6: Basic presentation + demo has problems

0-4: Cannot explain code + demo fails

Manifesto

Throughout the development of the FieldReserve application, our team maintained open communication and ensured that everyone contributed equally to the final product. To successfully transition our project to a REST API architecture, we divided the workload into complete vertical slices. Each member took full ownership of their respective module, developing the JPA entity, repository, service logic, REST controller, and the standalone frontend UI.

Our specific contributions are as follows:

Araz Hafez: Created the Field entity, managing the Field entity, FieldService, FieldRepository, FieldRestController, and the standalone frontend Field page.

Mohammed Adil: Created the Location entity, managing the Location entity, LocationService, LocationRepository, LocationRestController, and the standalone frontend Location page. Additionally, Mohammed served as the **Merge Manager**, handling integration testing, GitHub branch merging, repository cleanup, and database seeding.

Muhammed Rashid: Created the Reservation entity, managing the Reservation entity, ReservationService, ReservationRepository, ReservationRestController, and the standalone frontend Reservation page.

Osama Ahmad: Created the User entity, managing the User entity, UserService, UserRepository, UserRestController, and the standalone frontend User page.

Abstract

FieldReserve is a web application that aims at giving sports lovers and event organizers the ability to browse the various sport fields available and make reservations in a simple and conflict-free process. The application is built using the Spring framework and MariaDB; the goal is to make the process of booking fields simple and modern.

Problem Definition

At present, the process of booking sports fields for community use is complicated using scheduling platforms or phone calls. This could cause overlap in bookings or lack of clarity about the booking status of the field. The solution provided by FieldReserve is in formulating the availability of fields based on previous bookings and requesting bookings that would not result in overlapping problems. Requests can be made with chosen dates and fields.

Brief Description of Current Similar Solutions

Most of the available field management systems depend on bloated and complicated scheduling applications or use manual processes for making bookings using calls. The unique feature that sets FieldReserve apart from these commercial solutions is its specialized design for sports fields that offers simplicity in the user experience.

Technologies Used In Project

Backend: Java Spring Boot

Frontend: Simple HTML, CSS and Javascript (fetch/jQuery)

Database: MariaDB utilizing Spring JPA and Hibernate

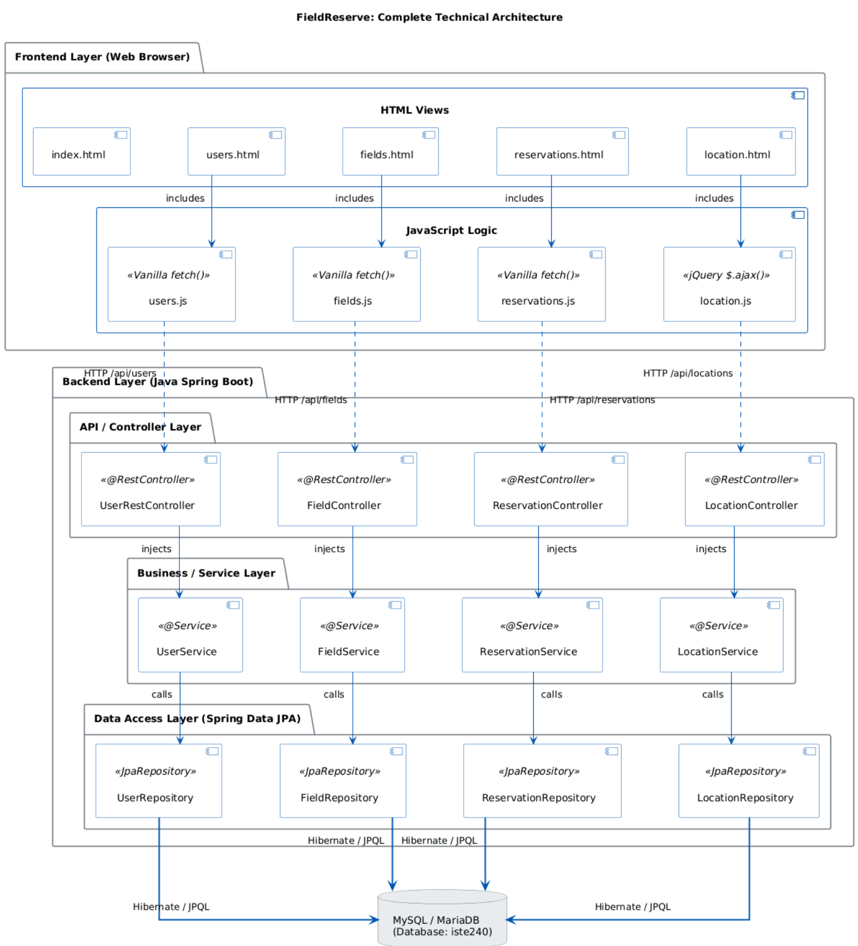
Tools:

GitHub for version control

Maven for dependency management

IntelliJ IDEA as the IDE

Technical Architecture



Key Website Features

User Interface: Landing page with navigation links, search bars, and responsive table layouts. Includes dynamic UI feedback such as success and error alerts. REST API Endpoints are reliably utilized for proper operation of the website.

Design Summary

The frontend was designed for high usability, utilizing Bootstrap to ensure responsive layouts. The application features a search bar, a table to view the contents, and a landing page that lets you navigate to other parts of the application. Users can submit, edit, and delete fields as needed within a simple and modern UI design.

Database design and summary

The MySQL database utilizes Spring Data JPA entity classes to define the data model. The primary entities include user, field, location, and reservation.
The JPA implementation includes custom repository methods such as findByEmailContainingIgnoreCase(), custom JPQL queries like findUsersByRole(), and update queries like updatePhoneById().

Repository

https://github.com/ma3067-AlwaysLearning/ISTE240.600-Team9